HHS3190M
HUMAN PHYSIOLOGY AND FUNCTIONAL ANATOMY FOR REHABILITATION SERVICES
HHS3190MJ
復康服務之人體生理學及功能性解剖學

Physiology L4: Cellular Energetics and Homeostasis
生理學 L4: 細胞能量學及體內平衡

(Sem 1 AY 26-27 26-27學年第1學期)

Outline大綱

- Basic concepts of cellular energetics and metabolism
  細胞能量學和新陳代謝的基本概念
- Nature and functions of enzymes
  酶的性質和功能
- Concepts of homeostasis
  體內平衡的概念
- Examples of homeostasis
  體內平衡的例子

Basic Concepts of Cellular Energetics and Metabolism
細胞能量學和新陳代謝的基本概念

- **Cellular Energetics 細胞能量學**:
  - Processes that a cell obtains, transforms, applies energy.
    細胞獲取、轉化、運用能量的過程。
  - These may include: 這些過程可能包括：
    - **Exergonic chemical reactions** that release energy
      釋放能量的**放能化學反應**
    - And **endergonic chemical reactions** that consume energy
      以及消耗能量的**吸能化學反應**
    - All these reactions are collectively known as **metabolism**
      所有這些反應統稱為**新陳代謝**

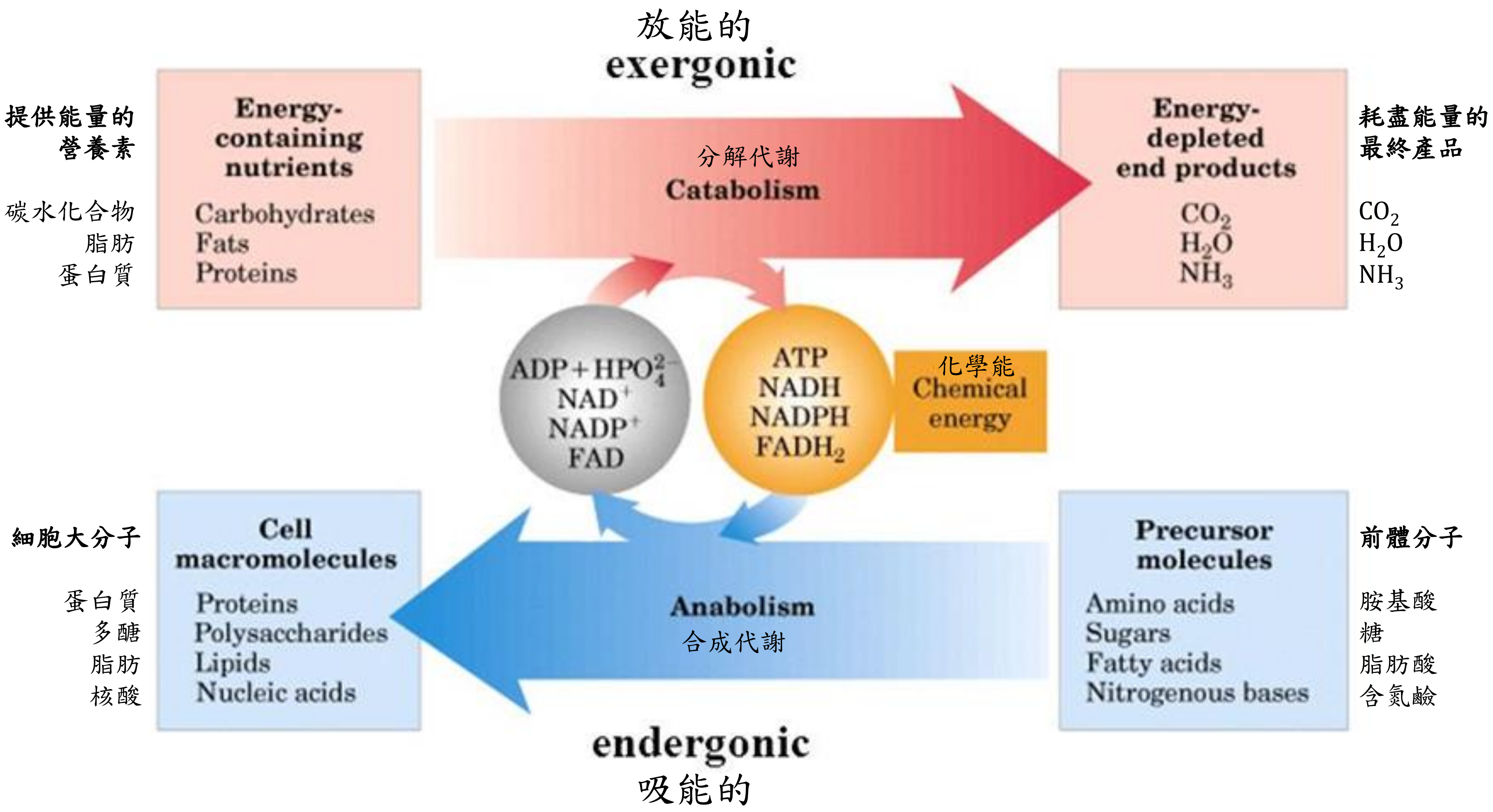
- **Metabolism 新陳代謝**:
  - Sum total of all biochemical reactions occuring in cell 細胞內發生的所有生化反應的總和

Basic Concepts of Cellular Energetics and Metabolism
細胞能量學和新陳代謝的基本概念

- Metabolic reactions are divided into 代謝反應分為:
- **Anabolism 合成代謝**:
    - Reactions that **use smaller molecules to build larger molecule**.
      E.g. using amino acids to make protein
      使用小分子構建大分子的反應，例如使用氨基酸製造蛋白質
    - **Cellular energy is used** in these processes. 這些過程中**消耗細胞能量**。
- **Catabolism 分解代謝**:
    - Reactions that **break down complex molecules to simple molecules**
      將複雜分子分解為簡單分子的反應
    - **Cellular energy is produced** in these processes.這些過程中**產生細胞能量**。
    - **Cellular respiration** is a group of catabolic reactions that **food fuels (E.g. glucose) are broken down to produce cellular energy**
      細胞呼吸作用是一組分解代謝反應，**食物燃料(例如葡萄糖)**被分解以產生細胞能量

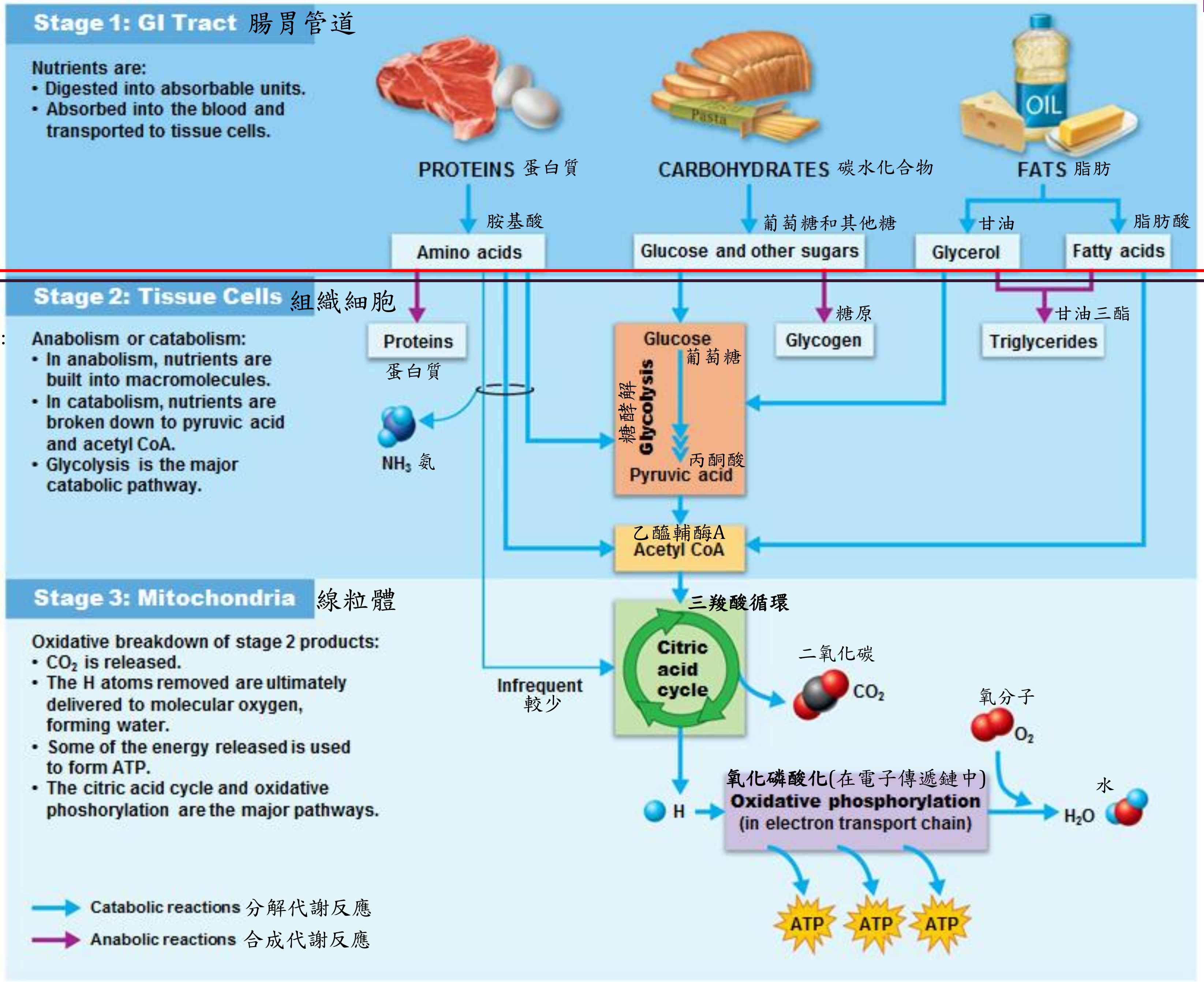
合成代謝和 分解代謝

Anabolism and Catabolism



營養成分是：

- 被消化成可吸收的單位
- 吸收到血液中並轉運到組織細胞中



合成代謝或分解代謝：

- 在合成代謝中，營養物質被合成為大分子。
- 在分解代謝中，營養物質被分解成丙酮酸和乙酰輔酶 A。
- 糖酵解是主要的分解代謝途徑。

第二階段產品的氧化分解：

- 二氧化碳被釋放。
- 去除的 氫原子最終被輸送到氧分子中，形成水。
- 釋放的一些能量用於形成 ATP。
- 三羧酸循環和氧化磷酸化是主要途徑。

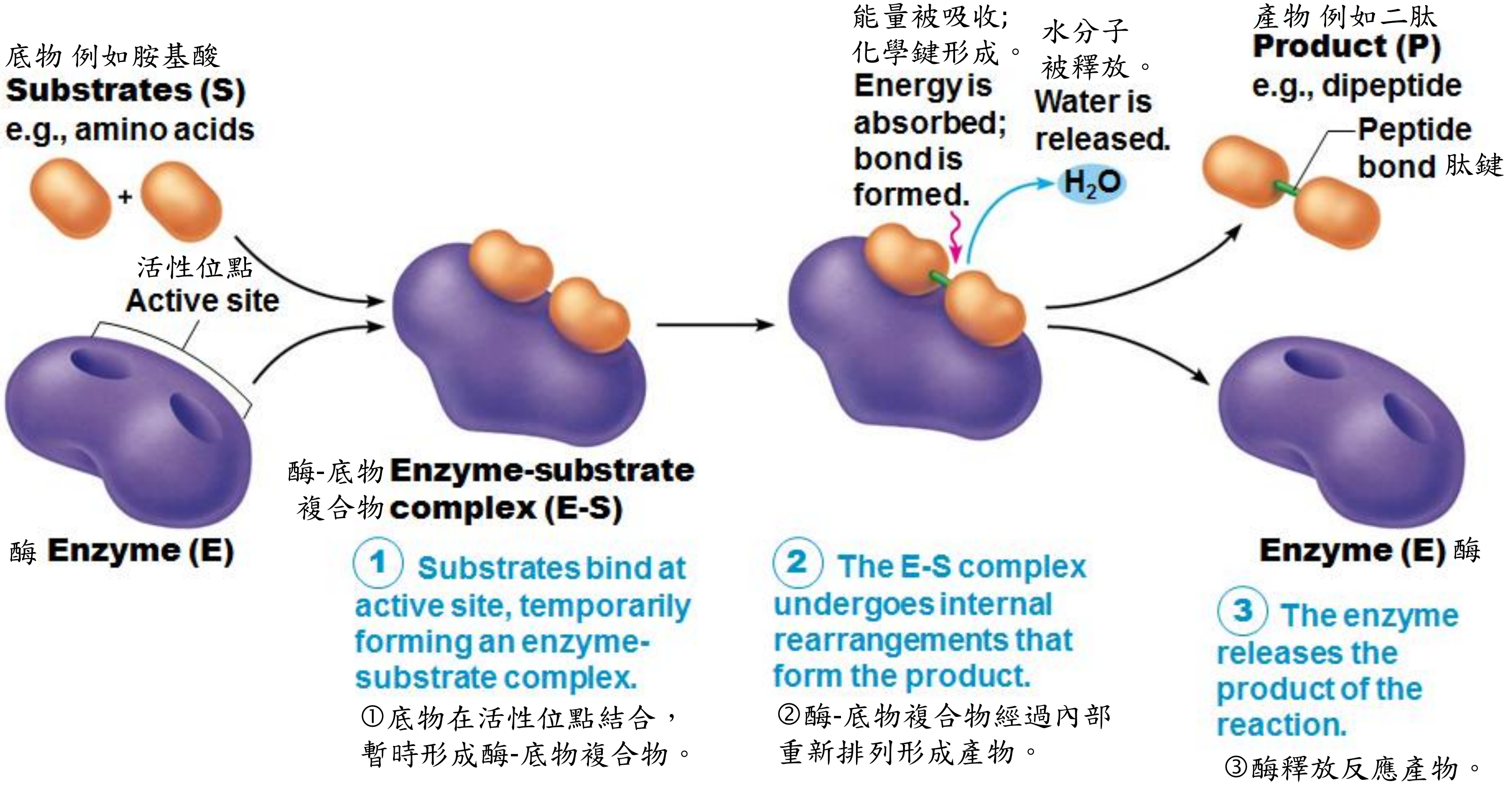
Nature and Functions of Enzymes
酶的性質和功能

- Most cellular reactions are organic reactions that occur slowly without catalysts. Therefore, there are many **enzymes** in human body to control rates of different reactions.
  大多數細胞反應是在沒有催化劑的情況下緩慢發生的有機反應，因此，人體內有許多**酶**可以控制不同反應的速率。

- **Enzyme酶**:
  - **Chemical nature**: Specific sub-class of **protein**
    化學性質：**蛋白質**的特定類別
  - **Active site**: **Special 3-dimension conformation** for binding of specific molecule **活性位點**：用於結合特定分子的**特殊三維結構**
  - **Function**: **Catalyze chemical reactions** of specific molecule
    功能：**催化**特定分子的**化學反應**
  - **Characteristic**: **Do not change** after the reaction **特點**：反應後**不改變**

Nature and Functions of Enzymes
酶的性質和功能

Three steps are involved in enzyme action 酶的作用涉及三個步驟：



Nature and Functions of Enzymes
酶的性質和功能

- The function of each enzyme is **specific** 每種酶的功能都是**特定的**
- Acting on **specific substrate** that fits with shape of active site
  用於適合活性位點形狀的**特定底物**

- Names of enzymes usually end in “**ase**” and are often named for the reaction they catalyze
  酶的名稱通常以“**酶**”結尾，通常因其催化的反應而得名
- Example例子：
- **ATPase** catalyzes**三磷酸腺苷酶** 催化
  - **ATP ↔ ADP + Phosphate 三磷酸腺苷↔二磷酸腺苷 + 磷酸基團**
- **Lipase** catalyzes **脂肪酶** 催化
  - **Triglyceride ↔ Monoglyceride + Fatty acids 甘油三酯↔甘油單酯 + 脂肪酸**

What is the function of following enzyme?
下列酶的作用是什麼？
(a) Amylase 澱粉酶
(b) Maltase 麥芽糖酶
(c) Aminopeptidase 胺肽酶

Nature and Functions of Enzymes
酶的性質和功能

- Enzymes are proteins. The structure (3-dimensional shape) of protein molecules can be changed by some factors. Thus these factors can alter the function of enzyme by changing the active site.
  酶是蛋白質，蛋白質的分子結構(三維形狀)可以通過某些因素改變，因此這些因素可以通過改變活性位點來改變酶的功能。

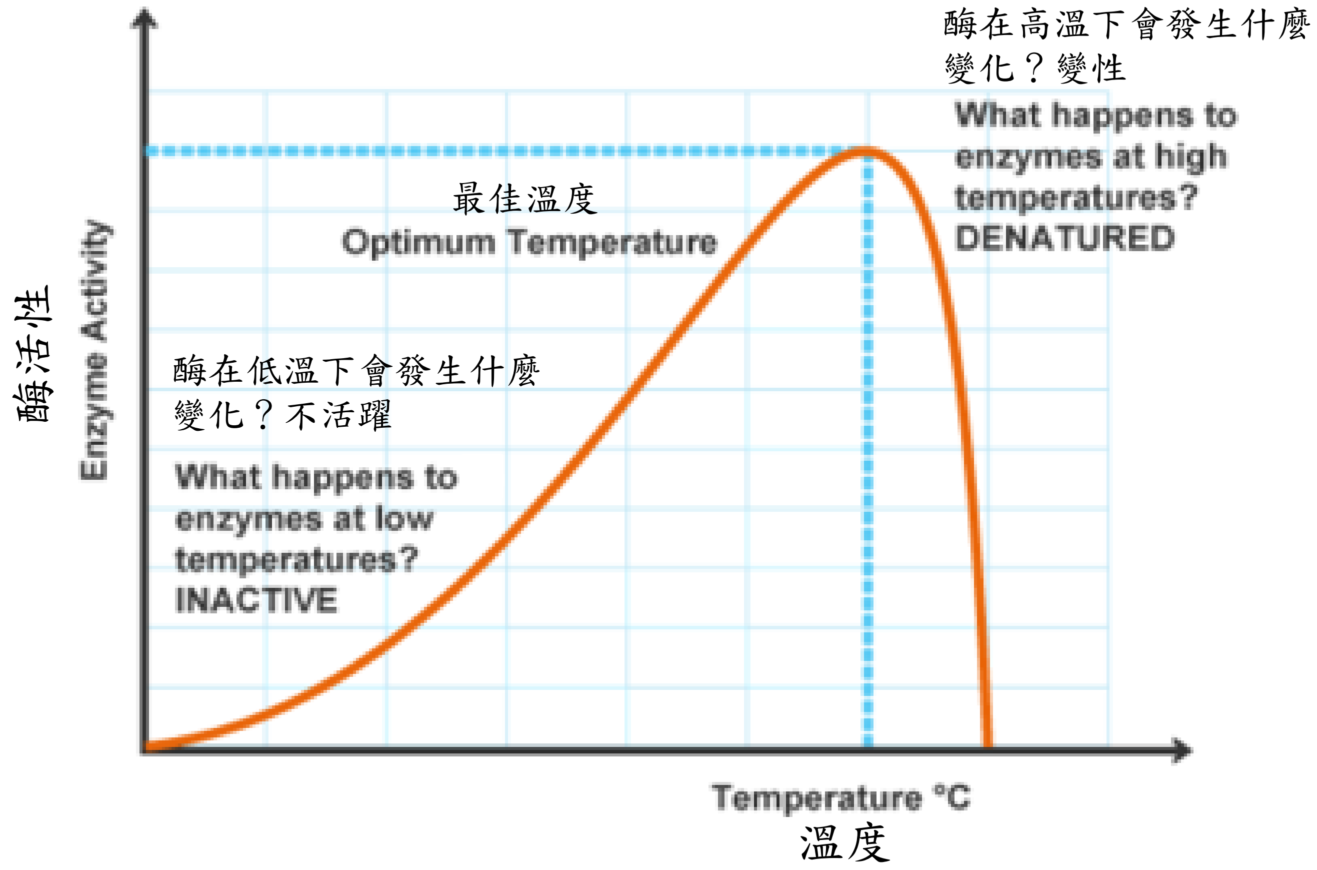
- Enzyme activity 酶活性:
  - the ability of an enzyme to catalyze reaction
    酶催化化學反應的能力
  - High enzyme activity → High reaction rate
    高酶活性→高反應速率

Nature and Functions of Enzymes 酶的性質和功能

- Factors that affect enzyme activity 影響酶活性的因素:
- (1) Temperature 溫度
- (2) pH level 酸鹼值
- (3) Presence of co-enzyme / co-factor 輔酶 / 輔因子
- (4) End-product inhibition 終產物抑制

Nature and Functions of Enzymes
酶的性質和功能

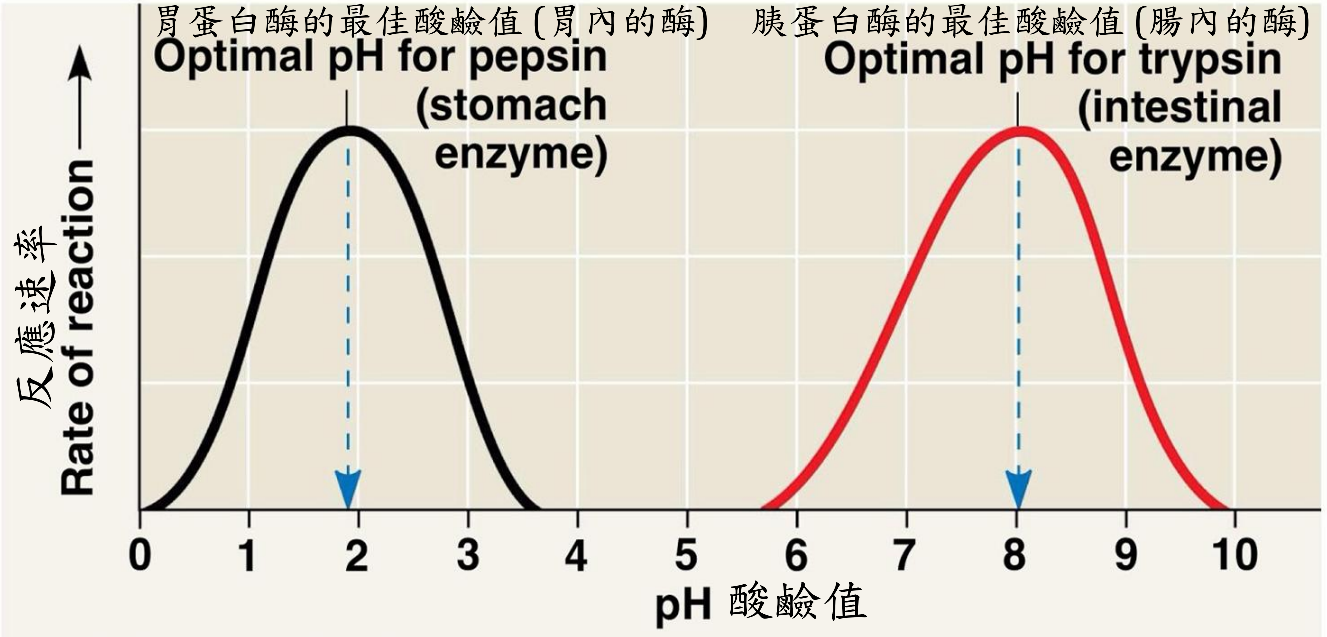
- Factors that affect enzyme activity影響酶活性的因素:
- **(1) Temperature溫度**
- When **temperature is low**, enzymes are **inactive**. **Low enzyme activity** leads to **low reaction rate**. When **temperature increases**, **higher enzyme activity** leads to **higher reaction rate**.
  當**溫度低**時，酶**不活躍**，**酶活性低**導致**反應速率低**。
  當**溫度升高**時，**較高的酶活性**會導致**較高的反應速率**。
- **Maximal reaction rate** is reached when temperature increases to **optimum temperature** (37-40˚C for most human enzymes).
  當溫度升高到**最佳溫度**(大多數人類體內的酶之最佳溫度為 37-40˚C)時，
  達到**最大反應速率**。
- When **temperature increases furtherly**, enzyme molecules are **denatured** to **decrease enzyme activities. Reaction rate drops** dramatically.
  當**溫度進一步升高**時，酶分子**變性**導致**降低酶活性**，**反應速率**急劇**下降**。


酶在高溫下會發生什麼
變化？變性
What happens to
enzymes at high
temperatures?
DENATURED
最佳溫度
Optimum Temperature
酶活性
Enzyme Activity
酶在低溫下會發生什麼
變化？不活躍
What happens to
enzymes at low
temperatures?
INACTIVE
Temperature °C
溫度

Nature and Functions of Enzymes
酶的性質和功能

- Factors that affect enzyme activity影響酶活性的因素:
- **(2) pH level酸鹼值**
- Most enzymes work within narrow range of pH level (**optimum pH**).
  大多數酶在狹窄的酸鹼值範圍內發揮功能(**最佳酸鹼值**)。

- **Extreme pH levels** (**Too alkaline** or **too acidic**) lead to **denaturation of enzymes** and thus **low enzyme activity** and **low reaction rate**.
  極端的酸鹼值(**太鹼** 或 **太酸**)會導致**酶變性**，
  從而**降低酶活性**和**降低反應速率** 。



(b) Optimal pH for two enzymes 兩種酶的最佳酸鹼值

pH level of gastric juice:
pH 1-3 (Acidic)

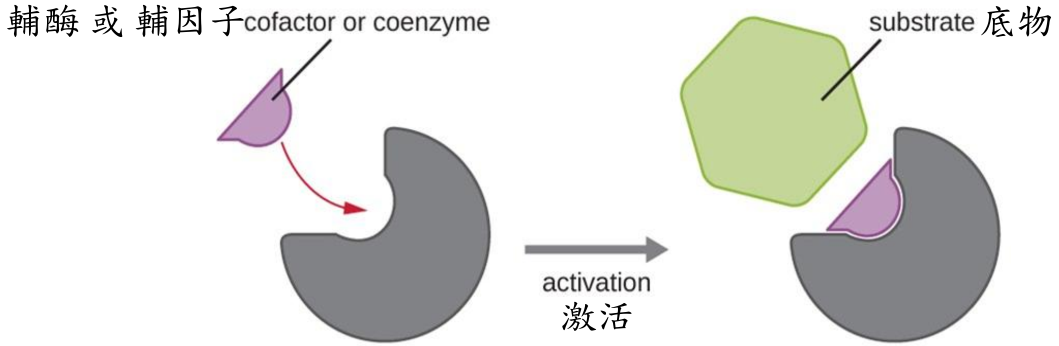
pH level of intestinal content:
pH 7-9 (Alkaline)

胃液的酸鹼值：
pH 1-3(酸性)

腸道內容物的酸鹼值：
pH 7-9(鹼性)

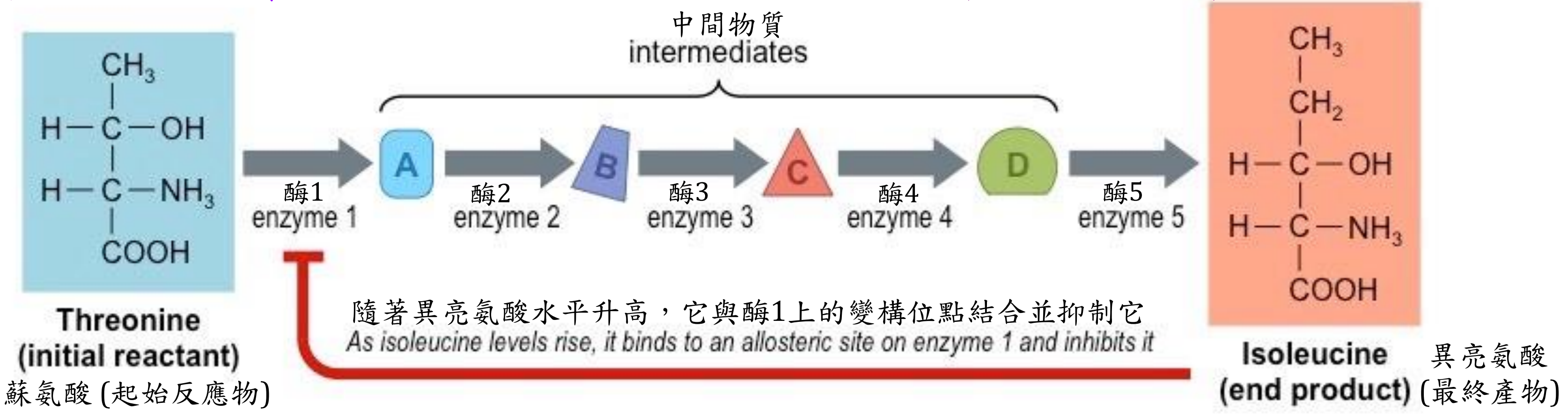
Nature and Functions of Enzymes
酶的性質和功能

- Factors that affect enzyme activity影響酶活性的因素:
- **(3) Presence of co-enzyme / co-factor 輔酶/輔因子的存在**
- Some enzymes need to combine with certain substances to become active
  有些酶需要與某些物質結合才能變得活躍
- If these substances are **organic molecules** (such as some water soluble vitamins), they are known as "**co-enzyme**"
  如果這些物質是**有機分子**(例如一些水溶性維生素)，則稱為 "**輔酶**"
- If these substances are **inorganic metal ions**, they are known as "**co-factor**"
  如果這些物質是**無機金屬離子**，則稱為 "**輔因子**"



Nature and Functions of Enzymes
酶的性質和功能

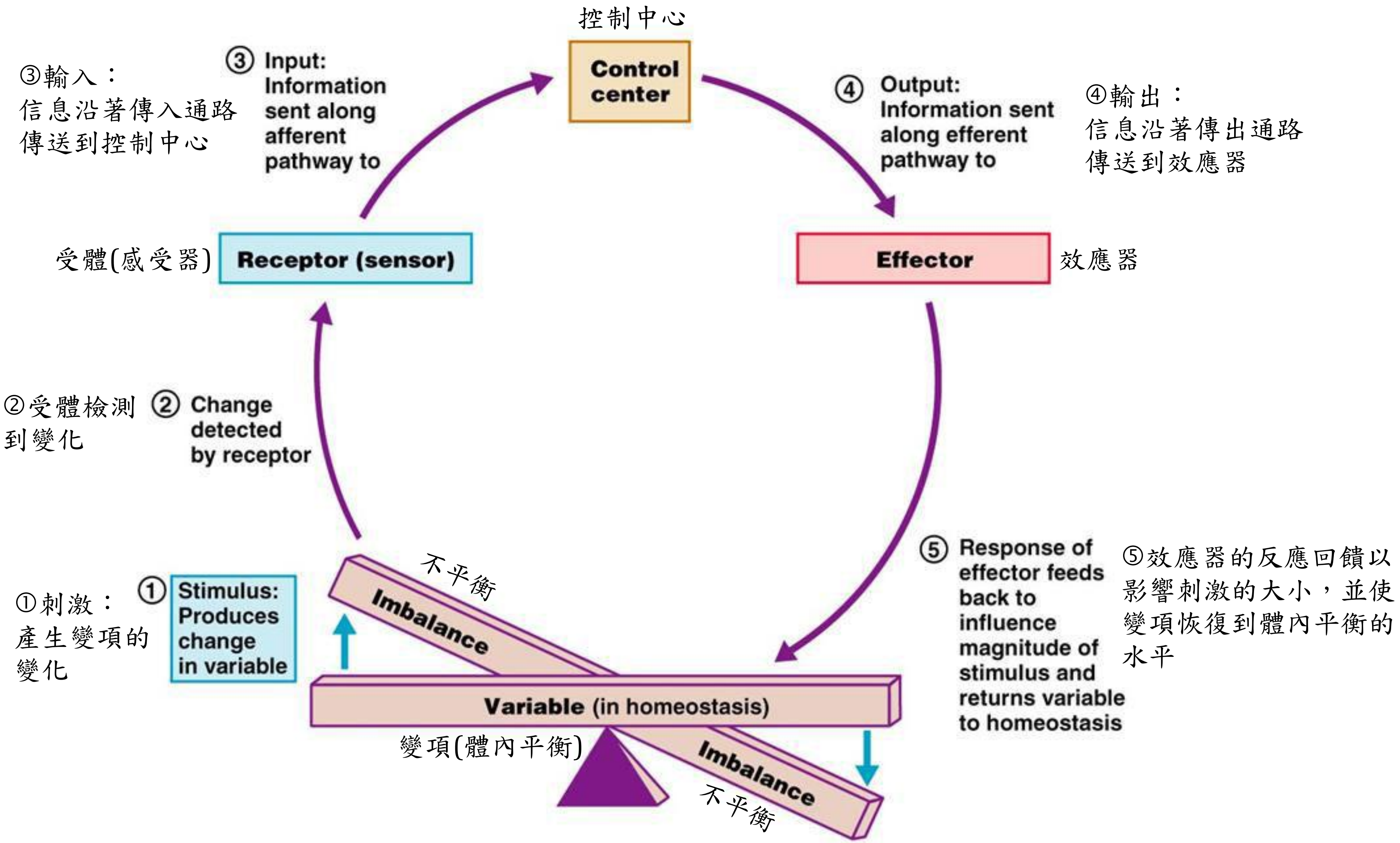
- Factors that affect enzyme activity影響酶活性的因素:
- **(4) End-product inhibition 終產物抑制**
- A **metabolic pathway** (a series of metabolic reactions) is usually controlled by **amount of end product** of the pathway.
  代謝途徑（一系列代謝反應）通常由該途徑的**最終產物量**控制。
- Large amount of end product **inhibit the enzyme of first reaction of the pathway** and thus decrease reaction rate.
  大量的最終產物會**抑制該途徑的首次反應酶**，從而降低反應速率。



Concepts of Homeostasis
體內平衡的概念

- The rate of chemical reactions in human body must be controlled to keep healthy body conditions. Therefore, human body must be able to **maintain stable and balance internal environment** by various methods. These ways are collectively known as "**homeostasis**".
必須控制人體內化學反應的速率以保持健康的身體狀況，
因此人體必須能夠通過各種方法來**維持內部環境的穩定和平衡**，
這些方式統稱為「**體內平衡**」。

- Common ways to maintain homeostasis維持體內平衡的常見方法:
- **(1) Negative feedback mechanisms 負反饋機制**
- **(2) Positive feedback mechanisms 正反饋機制**


控制中心
Control center
③輸入：
信息沿著傳入通路
傳送到控制中心
③ Input: Information sent along afferent pathway to
④ Output: Information sent along efferent pathway to
④輸出：
信息沿著傳出通路
傳送到效應器
受體(感受器)
Receptor (sensor)
Effector
效應器
②受體檢測
到變化
② Change detected by receptor
不平衡
Imbalance
①刺激：
產生變項的
變化
① Stimulus: Produces change in variable
Variable (in homeostasis)
變項(體內平衡)
Imbalance
不平衡
⑤ Response of effector feeds back to influence magnitude of stimulus and returns variable to homeostasis
⑤效應器的反應回饋以
影響刺激的大小，並使
變項恢復到體內平衡的
水平

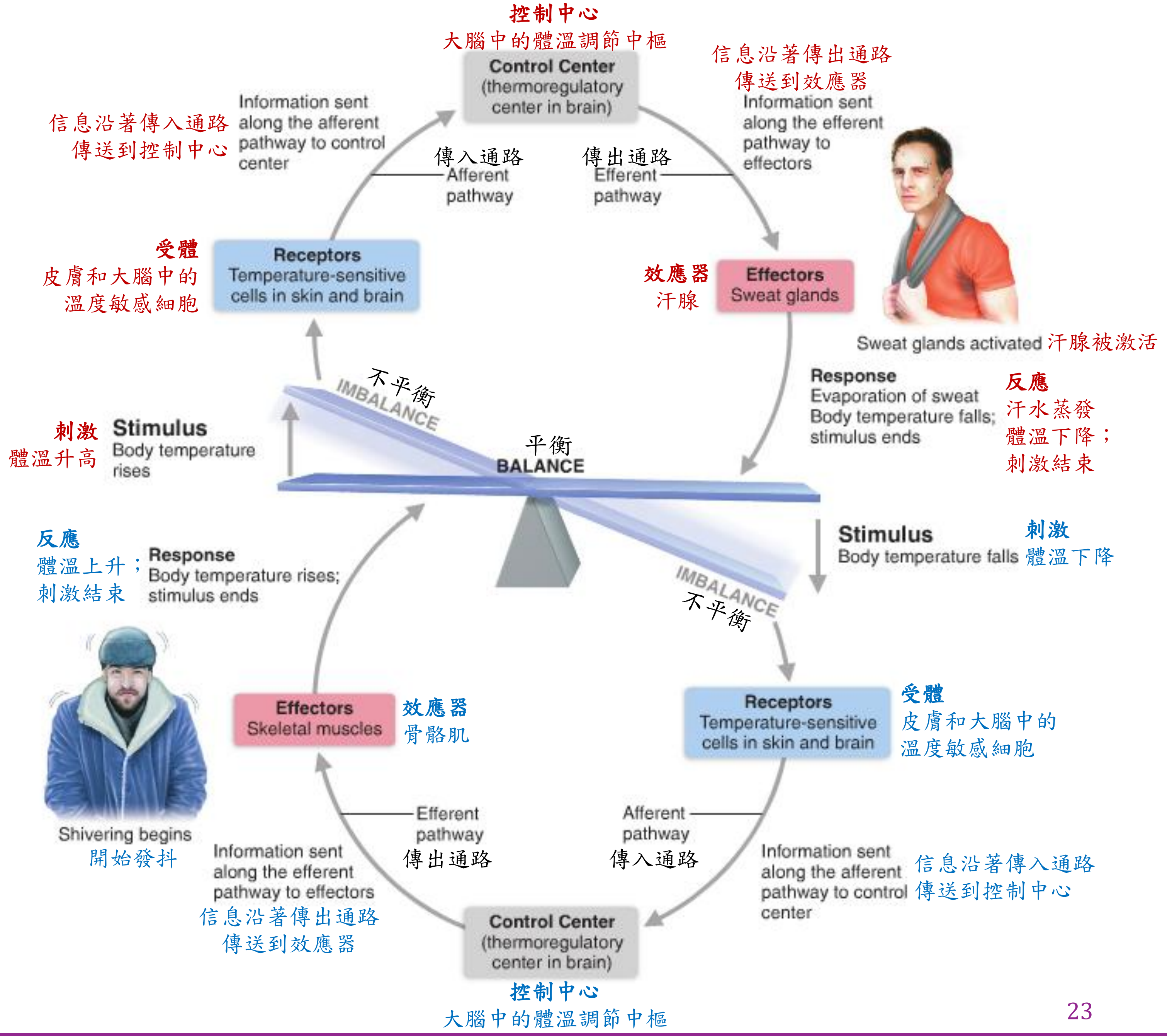
Concepts of Homeostasis
體內平衡的概念

- (1) **Negative feedback mechanism 負反饋機制**
- **Most homeostatic control methods** are negative feedback mechanisms.
  大多數體內平衡之控制方法是負反饋機制。
- In these methods, **outputs (responses)** are **opposite** with the **inputs (original stimuli)**.
  在這些方法中，**輸出(反應)**與**輸入(原始刺激)**相反。
- Therefore, the outputs (responses) reduce the inputs (original stimuli) to **keep within normal range**.
  因此，輸出(反應)減少輸入(原始刺激)以**保持變項在正常範圍內**。

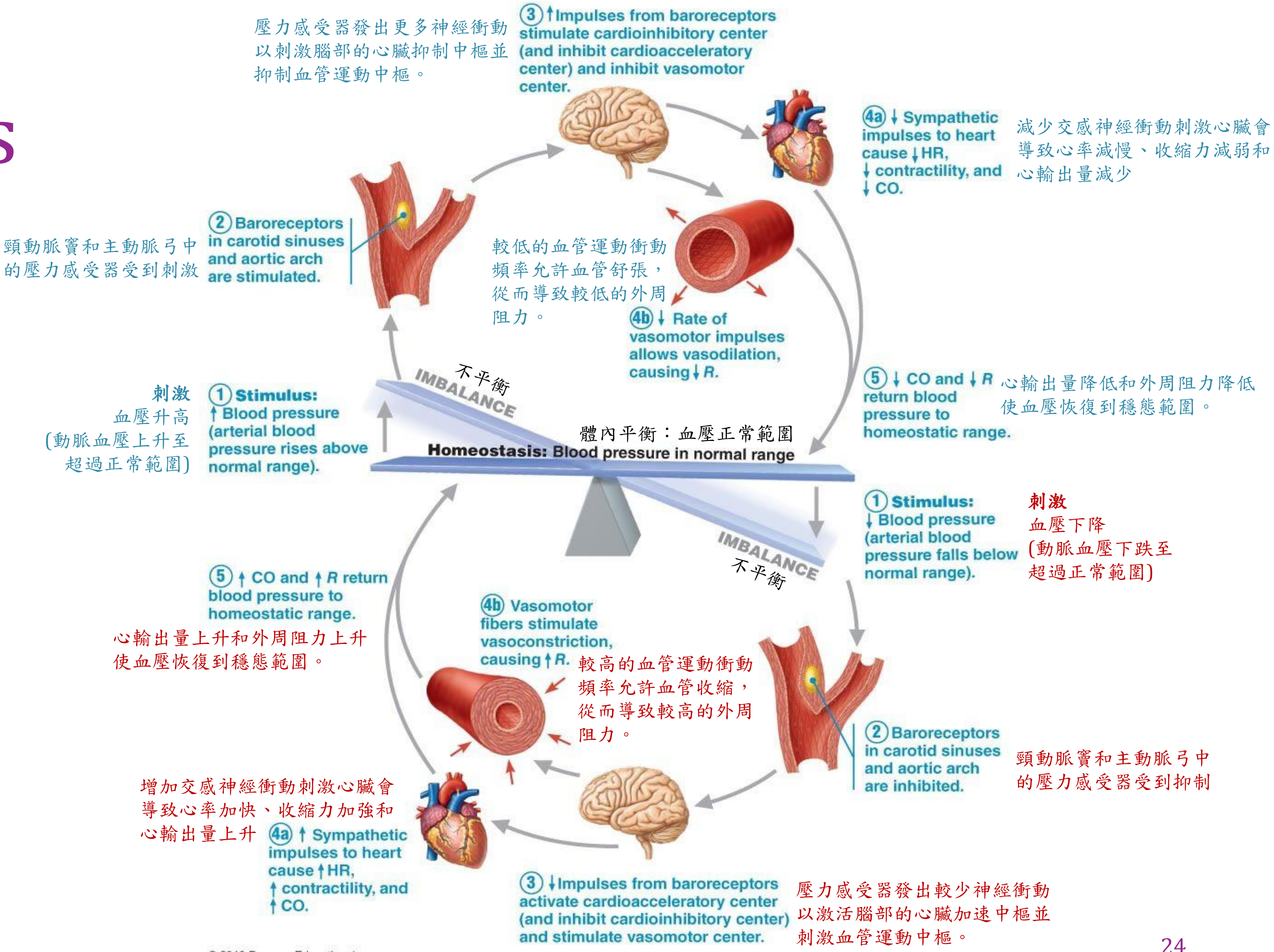
Concepts of Homeostasis
體內平衡的概念

- (2) **Positive feedback mechanism 正反饋機制**
- Sometimes use to control infrequent imbalance conditions
  有時用於控制不常見的不平衡情況
- Can be part of negative feedback control system
  可以是負反饋控制系統的一部分
- The **result/response (output)** has **same direction** as the **original stimulus (input)**.
  結果/反應(輸出)與**原始刺激(輸入)**具有**相同的方向**。
- Thus **output enhances input**.
  因此，**輸出增強了輸入**。

Examples of Homeostasis 體內平衡的例子



Examples of Homeostasis 體內平衡的例子

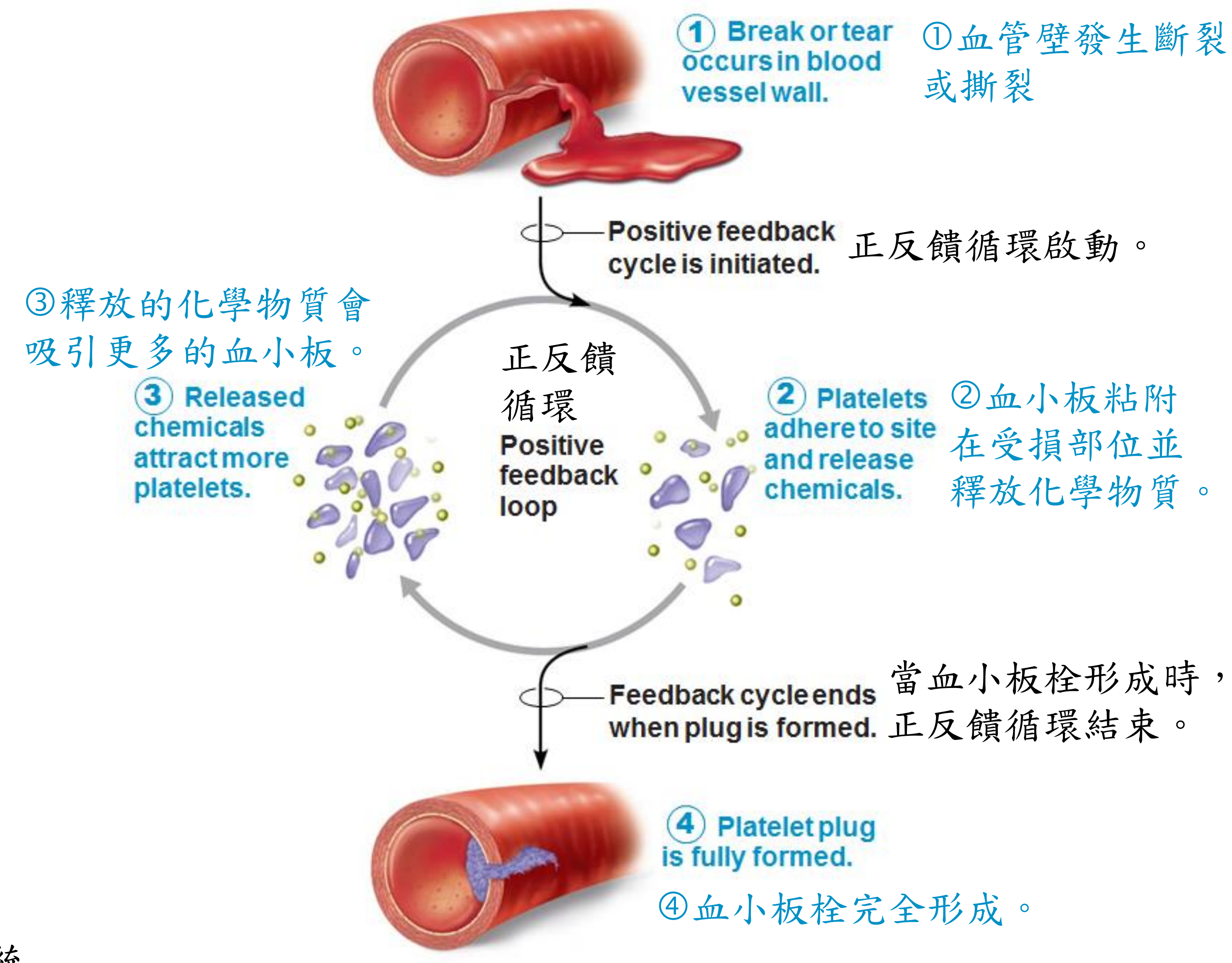


Examples of Homeostasis 體內平衡的例子

Negative feedback mechanism:
Stimulus: Bleeding
Responses: Stop bleeding

This is an example demonstrating "positive feedback cycle as a part of negative feedback control system".

負反饋機制：
刺激：出血
反應：止血
這是「正反饋循環作為負反饋控制系統的一部分」的例子。



Examples of Homeostasis
體內平衡的例子

Variable變項	Homeostatic range 穩態範圍	Remark 備註
Body Temperature 體溫	35.8-38.2ºC	
Blood pH 血液酸鹼值	7.35-7.45	
Blood Pressure 血壓	60-90 mmHg/ 90-140 mmHg	
Heart Rate 心率	60-100 bpm	
Fasting Blood Glucose 空腹血糖	3.9-5.6 mmol/L	WHO (2025) 世界衛生組織 (2025)

L4 Revision Exercise L4 複習練習

- Screen the QR code below to complete all questions in this revision exercise.
  掃瞄下面的二維碼以完成此複習練習中的所有問題。

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